

Please mind the doors — do interface metaphors improve the usability of voice response services?

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Results are presented from an experiment to measure objective and subjective user responses towards three implementations of an automated voice response telephone service, two embodying rich implementations of metaphors in the service messages including appropriate sound effects to enhance realism, a third using a non-metaphorical presentation of standard menu prompts. It was found that task completion rates, navigation of the service and attitudes were all improved with the use of metaphors.

1. Introduction

The standard menu style [1] used in many voice response services with TouchTone[®] input ('for A press 1, for B press 2, ...') is often recommended for its apparent simplicity and ease of use. It is ideal for the presentation of short lists and can significantly reduce interaction times for expert users who are familiar with the menu options and the use of 'key ahead'. However, it is also repetitive and as a result can be tedious and frustrating to use. Moreover, the essentially arbitrary character of its assignment of keys to service functions offers a cognitive and usability barrier for inexperienced users.

It has been argued that the introduction of a dominant metaphor deployed throughout the design of an interface may serve to make complex menu options more memorable and improve users' understanding of the system [2]. This should, in consequence, help longer-term retention of information about the structure of the service, accelerate the transition from novice user to expert and place fewer demands on users' concentration.

There is, therefore, a tension between the relative simplicity but arbitrary character of the standard menu style and the potential usability benefits to be derived from the systematic deployment of metaphors in the service interface. The experiment reported in this paper was designed to investigate this issue by examining the impact both on objective measures of performance, and on subjective evaluation of usability, for three implementations of an interactive voice response service using TouchTone

input from the user and pre-recorded service messages and prompts. One implementation employed a standard menu interface style (no metaphor); the other two were designed around different and contrasting metaphors.

2. A home shopping service

The voice response service used in this experiment was a hierarchically-structured home shopping service, as shown in Fig 1. There were two levels corresponding to categories of goods and specific items. To make a purchase, users were required to navigate from the top level to the level of individual items. At that point they could indicate how many of the items they wished to purchase. In order to leave the service users were required to select the 'exit' option from the top level menu.

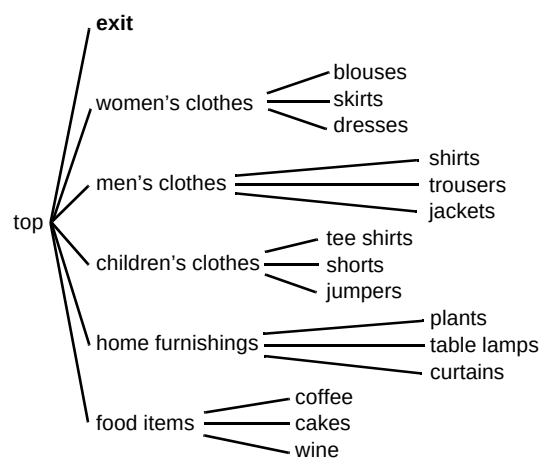


Fig 1 Structure of the database behind the home shopping service.

¹ TouchTone[®] is a registered trademark of British Telecommunications plc.

The standard menu style version of this service used information messages and prompts for the input of categories and items which were as literal and free from metaphorical associations with shopping as possible. It also offered users navigation options at each menu by making explicit reference to the key presses required to carry out specific actions. Examples of service messages and prompts are given in the first column of Table 1.

Table 1 Comparison of prompts in the three versions of the home shopping service.

Standard menu style (no metaphor)	Department store metaphor	Magazine metaphor
'For shirts press 1, for trousers press 2 and for jackets press 3.'	'Shirts are hanging up on your left, trousers are on the rail straight ahead of you, and jackets are over on your right.'	'You've got shirts in the first picture, trousers in the second, and jackets in the third'.
'To return to the list of groups of items, press star'.	'To return to the lift, press start'.	'To return to the contents page, press start'.
'Which category do you want?'	Liftman, 'Which floor do you want?'	'Which page would you like to turn to?'
'Category five, food items.'	'Going up! [sound effect: lift moving] floor five, food department.'	'[Sound effect: pages turning] page five, food items'

The other two versions of the service used different task domain metaphors to guide the design of message and prompt wording through the selection of appropriate figurative language. In one case the metaphor was that of a department store; in the other it was a magazine or shopping catalogue. Both metaphor versions of the service included sound effects to enhance realism. The metaphors were of wide scope [1] designed to provide users with a comprehensive view of the functionality of the service and its constraints. The mapping between keys on the telephone keypad and navigation within the service was established during the priming phase with each version of the service and was not referred to explicitly in the prompts or messages.

The department store metaphor induced a sense of moving between floors for different categories of goods and within floors for different items. Movement between floors was by means of a lift which was accessed using the star key on the telephone keypad. Audio effects were implemented to convey the sense of the lift going up or down and different voices were used in the lift and on the sales floors. All messages describing items were explicitly locational. For example, clothes were referred to as 'hanging up' to the left or right as the user left the lift, as illustrated in the second column of Table 1.

The magazine metaphor was equally rich and explicit but in this case navigation was represented as turning the pages of an illustrated catalogue. Categories of goods were organised as separate pages in the catalogue and individual items were described as numbered pictures on the page. Navigation through the magazine was via a contents page which was accessed by the star key. Sound effects for the turning of pages in a catalogue were implemented. The third column of Table 1 illustrates some of the messages used in this implementation.

The two metaphors were designed to manifest different relationships between the user and the service.

The prompts in the department store metaphor encouraged the feeling of a virtual three-dimensional space within which users themselves navigated (e.g. up or down between floors using the lift). By contrast, the prompts in the magazine metaphor encouraged users to think of themselves as remaining static while they turned pages as if reading a magazine or catalogue. In addition, the metaphors differed with respect to the semantics of the mapping between the telephone keypad and the menus. For example, in the case of the department store metaphor, the mapping was from the keys '1', '2' and '3' to 'left', 'straight ahead' and 'right' respectively. In the case of the magazine metaphor, the mapping was from the same keys to 'first picture', 'second picture' and 'third picture'.

3. Experiment procedure

The experiment employed a matched set design (with participants matched for age group and sex) to compare three groups of participants each using one version of the service. Volunteer participants with appropriate telephones were contacted at their homes from where they carried out the experiment. They were informed either that they would be using 'a telephone service that could be used to order goods' (standard menu group), 'a telephone service in the style of a department store which could be used to order goods' or 'a service in the style of a magazine'. No further details were supplied. Participants were then given three different shopping tasks to perform.

The emphasis throughout was on browsing and selection rather than activities such as the payment procedure. Following the last use of the service, participants completed an attitude and usability questionnaire. Throughout the experiment, objective data on task completion and responses (including instances of failure to respond to service prompts) were automatically logged.

A total of 60 participants took part in the experiment balanced for age group and gender.

4. Results

4.1 Task completion

The single most important feature of any automated service is that users can complete their tasks with it successfully. The primary objective measurement for this experiment was therefore task completion. Each task as presented to the participants was structured in two parts: it involved participants ordering a certain number of a named item in the store, e.g. 'three bottles of wine'. There were, therefore, five possible outcomes with respect to ordering an item using the service:

- no item ordered (transaction failure) (1),
- wrong number of the wrong item ordered (2),
- correct number but wrong item ordered (3),
- wrong number of the correct item ordered (4),
- number and item ordered correctly (5).

All participants succeeded in ordering an item each time they used the service, i.e. there were no transaction failures. There was only one case in each of the experiment groups of a participant ordering the wrong number of items and in each case it was the correct item (outcome 4 above). Figure 2 shows the proportions of participants for each experiment condition (i.e. interface style) and each use of the service who completed the task successfully (outcome 5 above). Given the small number of instances of outcomes 1, 2 and 4, the error levels shown in Figure 2 are almost entirely attributable to participants ordering the correct number of the wrong item (outcome 3).

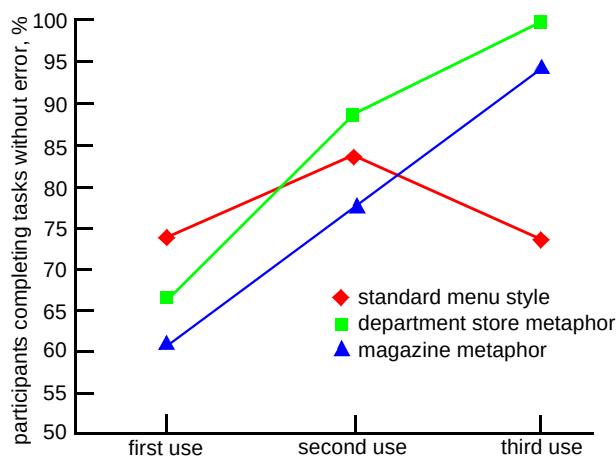


Fig 2 The proportion of participants completing tasks without error.

McNemar binomial distribution two-tailed tests showed that between the first and third exposures there was a significant improvement in task completion accuracy for participants using the department store metaphor ($p = 0.03$) and an improvement approaching significance for users of

the magazine metaphor ($p = 0.07$). Chi-Square tests showed that for the third use there were significant differences between the interfaces with metaphors and the interface without metaphors ($p = 0.02$). The differences between the experimental groups in the first and second exposures were not significant.

Clearly, participants who used the interfaces designed around specific and explicit metaphors learned how to use the service more successfully over the three exposures than those using the service employing the standard menu style. Although success rate was slightly better with the standard menu version for first time users, it was not significantly so. This suggests that the use of metaphors did not negatively impact performance with the service for first time users, although it is possible that the more open prompt style of the metaphor versions may have depressed performance on the initial call. The use of metaphors clearly enhanced performance with the service relative to the literal version for users with previous experience with the same service. This measure did not, however, separate the two metaphors — both appeared equally effective in improving user performance.

4.2 Exiting the service

In order to leave the service participants were required to enter the sequence '* 0' on the telephone keypad - pressing the star key returned them to the top level from which '0' was the exit option. It was therefore necessary in all three versions of the service for participants to remember to press '*' before pressing '0' to exit. Table 2 shows that all the participants using the department store metaphor first pressed '*' (to go to the lift) before pressing '0' (to leave the store). This suggests that the store and lift metaphor was a very effective model for the requirement to change modes.

Table 2 Occurrences of failing to press '*' before '0' when attempting to exit the service.

	First use	Second use	Third use
Department store metaphor	0	0	0
Magazine metaphor	5 (5 participants)	3 (3 participants)	0
Standard menu style	0	1	7 (4 participants)

More participants made this error with the magazine metaphor, although an improvement was seen over the course of the three uses of the service — by the third use there were no exit errors. Participants who made this error reported that it was not obvious during the first use that the magazine could only be closed from the contents page

(accessed by pressing star). However, by the third exposure they had successfully learned the correct procedure.

The frequency of this error on the first use with the magazine metaphor suggests that the lack of a 'natural' one-to-one mapping between the elements of the magazine metaphor and the service functionality (in the real world, a book can be shut at any page) created some confusion. However, the fact that there were no errors of this type by the third use of the service strongly suggests that participants had adapted their mental models of the mapping between the metaphor and the service appropriately, a fact supporting Carroll and Mack's observations [3] that people are not necessarily confused by mismatches between metaphorical and target domains. They in fact argue that fuzziness and incompleteness in the metaphor-to-system mapping may promote active and creative thought on the part of the user.

For the standard menu style there were no occurrences of this error in the first use but by the third use there were more errors than in the other two experimental groups. One explanation for this unexpected finding is that although the instructions were clear on how to exit from the service and involved the same procedure as the metaphor versions, the lack of a meaningful metaphor meant that participants did not have the advantage of being given a framework with which to understand and remember this aspect of the system. An alternative explanation is that as participants used the service more, they used 'key ahead' more and listened less carefully to the service prompts.

4.3 No user response

Figure 3 shows that the rate of failure to respond to prompts fell in all of the groups across the three exposures. The magazine metaphor group had the fewest instances of failure to respond overall.

Failures to respond using the telephone keypad in both the metaphor conditions were largely the result of participants speaking to the service at beginning. By design, the dialogue prompts of the two metaphor versions of the service did not refer explicitly to the need to press keys on the telephone keypad until error recovery was invoked (for example, first prompt of the department store version: 'Which floor do you want?'). In contrast, the standard menu prompting style was explicit about this (for example, 'For men's clothes press one'). As a result, approximately 50% of the participants using the metaphor versions spoke to the service at the first prompt while only two of the twenty standard menu users did so.

Typical responses from department store metaphor users included: 'I would like floor four please' and 'Floor three please'. Responses from the magazine metaphor users included: 'Can I have page four', 'Page four please'.

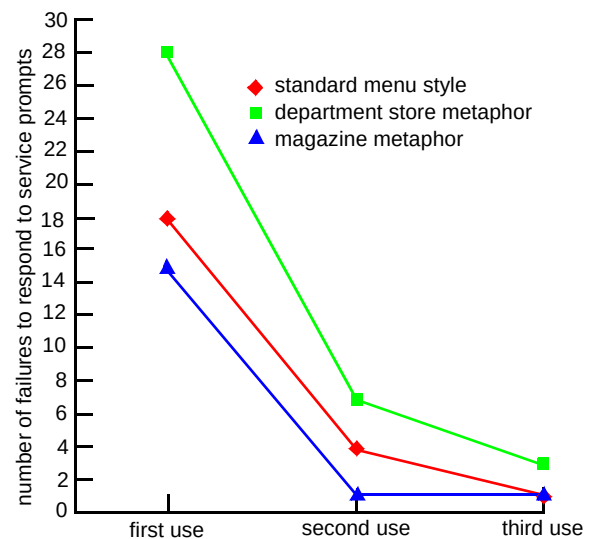


Fig 3 Failures to respond to service prompts by use and experimental condition.

4.4 Subjective measures

The usability questionnaire was designed to evaluate the significance of factors important in the assessment of usability. These include the quality of the interface performance (e.g. perceived efficiency), the cognitive effort required to use the service and the stressfulness of the experience (e.g. the degree of concentration required, the frustration of the experience), the conversational model implied by the interface, the fluency of the interface and its transparency. The questionnaire was cast in a Likert equal-interval format where each item consists of a stimulus statement accompanied by a seven-point response scale ranging from 'strongly agree' to 'strongly disagree' with a mid neutral point. The 22 stimulus statements are listed in Table 3.

Figure 4 shows the overall mean scores for the three versions of the service. T-tests revealed that there was a significant difference between the department store and standard menu groups' overall mean rating of the service ($p = 0.005$) but the difference between the magazine and standard menu groups was not significant.

Responses to individual usability and attitude statements show interesting variations. Three groupings of usability dimensions and attitudes are of special interest:

- dimensions which show no significant difference between the three versions of the service,
- dimensions for which the two metaphor versions are significantly different from the standard menu version but where the metaphor versions do not differ from each other,
- dimensions where the two metaphor versions differ from each other.

Table 3 Stimulus statements for the Likert usability questionnaire.

I had to concentrate hard when using the service.	I liked the wording of the messages.
I didn't enjoy using the service.	It was always obvious how to find what I wanted.
I found the messages repetitive.	I thought the voice was/voices were very clear.
I thought the service was efficient.	I thought that it took too long to order an item.
I thought that the service was easy to learn to use.	I felt in control while using the service.
I thought the service was friendly.	I found the service confusing to use.
I found it easy to make my responses with the telephone keypad.	I felt frustrated when using the service.
I sometimes felt lost while using the service.	I found the service easy to use.
I thought the messages were easy to follow.	It would have been useful if I could have requested more help from the service.
It wasn't always clear which key I needed to press.	I didn't like the voice/voices.
I thought the service was fun.	I thought that some of the messages were too complicated.

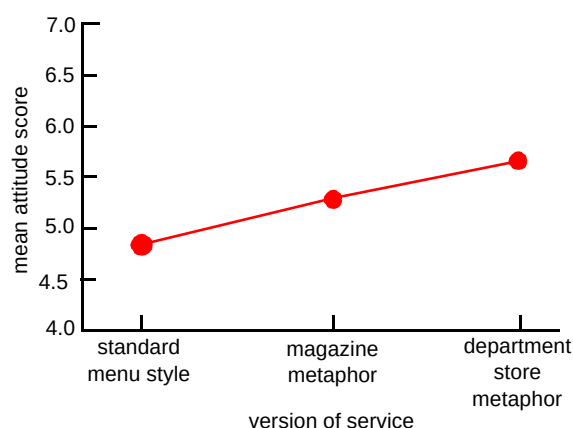


Fig 4 Overall mean rating of attitude statements.

The combination of these indicates a great deal about the relative usability of the three versions of the service and the (possibly different) contributions made by the metaphors to attitudes towards the service.

There were no significant differences between the three versions for factors such as the clarity of the voice, ease of use of the telephone keypad, overall ease of use of the service, ease with which the service procedures could be learned and the perceived length of time it took to complete the tasks. Some of these similarities were expected since, for example, the voice was the same in all three versions of the service. It is interesting to note, however, that there was no difference in perceived ease of use between the metaphor

and standard menu versions of the service — adding a metaphor to the standard menu style does not necessarily improve the perceived ease of use of the service, nor does it appear to users to make the service faster or slower.

Both metaphor versions were considered to be more fun to use than the standard menu version. Participants experienced a greater degree of control with the metaphor versions, felt they needed less help, liked the voice much more and, overall, enjoyed the experience more than with the standard menu version. These results are encouraging insofar as they lend support to the idea that the introduction of an explicit metaphor into the service interface makes the experience more pleasant and therefore probably contributes to the overall increase in perceived usability of the service.

It was also found that the two metaphor versions differed from each other with respect to certain usability factors. The magazine metaphor did not score significantly better than the department store on any of the measured dimensions. The department store, however, was felt to be less frustrating to use, required less concentration, was more friendly, produced less feeling of confusion and less feeling of being lost than the magazine metaphor. The department store metaphor was more effective than the magazine metaphor in reducing the stressfulness and cognitive loading for most participants.

4.5 Gender differences

There were no significant attitude differences between genders for users of the standard menu style or the department store versions of the service. There were, however, significant differences in attitude between genders towards the magazine metaphor version, with women having a significantly more positive attitude than men. It is possible that the women found the style of verbal presentation more congenial than the men (e.g. 'You've got trousers in the 1st picture, shirts in the second ...') and that they may have had more experience and knowledge of magazines and ordering goods from catalogues. This difference — which merits further investigation — suggests that simple generalisations across different user populations about the effectiveness of metaphors in the interface should be avoided. The three questionnaire items which showed significantly different attitudes (ease of use, messages easy to follow, and some messages too complicated) support this line of argument.

5. Conclusions

In this experiment, task completion rates were significantly improved with the use of the metaphors; navigation of the service (exemplified here by the procedure for exiting the service) was also improved. The substantial reduction in the failure to respond to service prompts which was found

with repeated use of all three versions of the service was as significant with the metaphor versions as with the standard menu version (the introduction of a metaphor into the service interface did not adversely influence this aspect of user behaviour). Lastly, subjective attitudes were significantly more positive towards the department store metaphor version of the service.

Overall, the experiment clearly demonstrated that the two metaphors used in this experiment produced an effective improvement of the speech interface. The fact that female participants had a more positive attitude towards the magazine version than male participants is a strong indication that the type of metaphor used should be tailored to accommodate the characteristics of the targeted users.

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